



DNA & DATA SCIENCE CORES

at Delaware State University

SEQUENCING THE FUTURE

DSU dsDNA Core uafR Undergraduate Research Training Program

Instructor and Mentor Guide

Facilitating a 10-week reproducible chemical research internship

Guide Purpose

This companion guide provides program setup, pacing, review questions, coaching guidance, assessment evidence, and release procedures. Student tasks, worksheets, rubrics, and the complete 70-day schedule remain in the student manual.

DSU dsDNA Core

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github.com/castrattonDSU/uafR

Curriculum and institutional-mark notice: see `training/PUBLICATION_NOTICE.md`.

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About This Guide

This guide accompanies the *DSU dsDNA Core uafR Undergraduate Research Training Program: Student Manual*. It is written for instructors, research mentors, and program staff who are responsible for preparing the learning environment, reviewing student evidence, and deciding when a project is ready to advance through the program gates.

What Is Shared Publicly

The public curriculum includes the student manual, this generic instructor guide, training scripts, worksheets, rubrics, simulated examples, and installation tools. This transparency allows students to understand how their work will be evaluated and allows other research programs to inspect the curriculum before adopting it.

What Belongs in a Local Supplement

The public guide does not define institution-specific employment rules, compensation, schedules, accommodations, emergency procedures, unpublished project details, credentials, or restricted-data locations. Before the program begins, those items should be documented in a locally approved supplement and reviewed with every participant.

Companion Documents

- The student manual contains the complete 70-day curriculum, labs, readings, project tracks, worksheets, rubrics, and student setup guide.
- The package README and pkgdown site contain current function-level documentation and species-first plant phytochemistry examples.
- The offline student bundle provides a fixed package archive, the student manual, scripts, integrity checks, and an acceptance test.
- Repository workflows verify that public manuals build and that the offline training path still runs against the current package.

Support Boundary

The curriculum teaches evidence-aware analysis; it does not replace laboratory safety training, responsible-conduct requirements, data-governance approval, chemical identity confirmation, statistical consultation, or project-specific scientific supervision.

1 Program Implementation Guide

1.1 Program Philosophy

This program treats undergraduate interns as early-career researchers. Students are not expected to arrive knowing R, chemistry, statistics, mass spectrometry, or manuscript writing. They are expected to learn how to make their work traceable, honest, reproducible, and useful. The curriculum is intentionally full-day: research skill develops through repeated cycles of trying, inspecting, documenting, revising, and explaining.

1.2 Pre-Program Setup Checklist

Before Week 1

- Prepare the current public uafR installation route or a verified offline student bundle.
- Confirm that the bundle includes installation instructions, preflight checks, acceptance tests, training scripts, and this manual.
- Prepare a case dataset and short chemical list for students without immediate research data.
- Decide where students will store project folders and which locations may contain private or unpublished data.
- Prepare printed or digital copies of the Project Brief, Chemical Intake Table, Database Evidence Map, Daily Research Log, and Weekly Progress Review worksheets.
- Schedule weekly review checkpoints and final presentation time.
- Identify which public database exercises can use saved examples if network access or live services fail.
- Create a list of program contacts for R help, chemistry interpretation, data privacy, and presentation review.

1.3 Local Program Supplement

Before enrollment or employment begins, provide a short locally approved supplement that states the actual calendar, expected weekly hours, how weekend tasks count toward those hours, compensation

or credit terms, holidays, supervision, accommodations, laboratory safety requirements, emergency contacts, and data-access rules. The 70-day curriculum defines learning activities; it does not supersede institutional employment, academic, accessibility, or safety requirements.

Keep unpublished project names, restricted-data locations, credentials, and internal scheduling details in the local supplement or another approved controlled location. Do not add them to the public manual or student bundle.

1.4 Daily Check-In Script

The daily check-in is short and artifact-centered. It can be used at the start or end of a workday.

1. What did the student produce since the last check-in?
2. Where is the file, table, figure, worksheet, or code object saved?
3. What evidence supports the main thing the student thinks they learned?
4. What is uncertain, missing, broken, or overclaimed?
5. What will be produced by the next check-in?

1.5 Weekly Mentor Review Script

Weekly review should inspect the product ladder artifact rather than reviewing the whole project from scratch.

1. Confirm the week's primary deliverable and file location.
2. Ask the student to explain the research question in one minute.
3. Ask the student to show the source evidence behind one important claim.
4. Ask the student to show one reproducibility check or rerun result.
5. Ask the student to name the strongest limitation.
6. Decide whether the next week can proceed, proceed with a reduced scope, or pause for revision.

1.6 Decision-Gate Review Questions

Gate	Review questions	Evidence required
Scope gate	What chemical set is being studied? What claim is out of scope?	Project brief, risk note, chemical intake draft.
Reproducibility gate	Can the project be opened and rerun from its root?	README, setup script, session info, clean folder structure.

Gate	Review questions	Evidence required
Evidence gate	Which sources returned useful data and which did not?	Validation summary, source coverage, diagnostics, saved result object.
Variable gate	How did source evidence become analysis-ready variables?	Grouping dictionary, trait evidence audit, matrix settings.
Interpretation gate	Which claims survive evidence review?	Claim-evidence-limitation memo, figures, literature table.
Handoff gate	Can someone continue the project without the student present?	Final audit, continuation note, re-run status, archive path.

1.7 Expected Student Answers by Program Stage

Stage	Strong answer	Answer needing revision
Question design	Names system, chemical set, comparison, evidence type, and limitation.	"I want to see what chemicals are important."
Identifier cleanup	Preserves original names and documents cleaned query names and ambiguity.	Original names are overwritten or difficult compounds disappear.
GC-MS workflow	Describes match factor, retention time, base peak, area, and tentative identity caveats.	"The software identified the compound."
Database enrichment	Starts with validation and source coverage.	Jumps directly to the most exciting pathway or bioactivity term.
Trait design	Names source table, field, extraction rule, allowed values, confidence, and caveat.	Uses broad sentences as categories.
Visualization	Explains which question the figure answers and which values are missing.	Shows a figure because it looks interesting.
Literature	Uses papers to refine interpretation and limitations.	Searches only for papers that support a desired conclusion.
Final report	Methods, results, and limitations are reproducible and source-backed.	Report reads like a story without an evidence trail.

1.8 Common Coaching Moments

Pattern	Useful program response
Project is too broad	Reduce to one chemical set, one comparison, and one main evidence family.
Student is stuck on installation	Use the private bundle path, run preflight checks, and switch to case-data reading while installation is repaired.
Student has weak R background	Keep chemical set small, require script comments that explain intent, and prioritize clean reruns over advanced analysis.
Student has no chemistry background	Use identifier ambiguity drills and claim-boundary examples before interpreting traits.
Student has strong background and needs more challenge	Add sensitivity analysis, reviewer response, literature conflict, or a reusable audit/plotting function.
Live service fails	Use saved result objects or offline examples and document that live access was unavailable.
Matrix is too wide	Return to the grouping dictionary and remove variables that lack project relevance or source support.
Writing overclaims	Rewrite the claim using evidence type, source, and limitation.

1.9 When To Slow Down

The program should slow down when a student cannot explain the question, cannot find the files they produced, ignores source coverage, interprets before validation, uses absolute paths, treats database text as proof, or cannot explain what a figure shows. Slowing down means reducing scope and improving evidence quality, not abandoning the project.

1.10 When To Accelerate

The program can accelerate when the student can rerun the workflow from a clean session, source coverage and validation are documented, the grouping dictionary is clear, figures have source-backed captions, and the student can explain what evidence would change their interpretation. Acceleration should add depth: sensitivity checks, literature conflict analysis, stronger figures, or reusable functions.

1.11 Running With Real Data, Case Data, or No Data

Mode	Best use	Required adjustment
Real student data	Strongest ownership and publication potential.	Add data privacy notes, raw-data preservation, and realistic scope limits.

Mode	Best use	Required adjustment
Program case data	Best for teaching a consistent workflow.	Make clear that scientific claims are about the case exercise, not new discovery.
No available chemical data	Best for students entering early or from communication tracks.	Use a curated chemical list, database evidence atlas, methods evaluation, or teaching module.
Mixed mode	Best when real data are messy.	Train on case data first, then apply only reviewed steps to real data.

1.12 Protecting Unpublished and Private Data

Private and unpublished data require deliberate handling. Student projects should not contain tokens, passwords, private collaborator files, raw data with restricted identifiers, or uncontrolled copies of unpublished sample metadata. Public-facing products should use generalized descriptions unless sharing has been approved. Reproducibility can be preserved with documented access rules, checksums where appropriate, case-data substitutes, and clear separation of public scripts from private data.

1.13 Database Etiquette

Live database access is deliberately conservative:

- Start with one or two compounds before scaling.
- Use caching for repeated work.
- Use throttling for broad queries.
- Save RDS result objects so students do not repeat identical requests.
- Use offline saved examples when teaching in rooms with unstable internet.
- Treat service failures as workflow realities, not as student failures.

1.14 Evaluating Final Packages

Final review uses three questions:

1. Can the package be understood from the README, report, and continuation note?
2. Can the figures and tables be traced to scripts, saved objects, source evidence, and literature?
3. Are claims scientifically restrained by tentative identity, source coverage, missingness, and evidence type?

Packages that pass these questions can be archived even if they contain negative or inconclusive findings. Inconclusive projects are valuable when they document why the evidence was insufficient and what should be done next.

1.15 Archiving Projects

The final archive should include the final project folder, report, presentation, saved uafR result object, validation outputs, source coverage report, trait matrix, grouping dictionary, figures, research log, session information, final package audit, and continuation note. Archive names should include student initials or project code, year, and short title. Restricted data should remain in controlled storage with a note explaining access requirements.

1.16 Recommended Grading Weight

Component	Weight	Notes
Weekly deliverables and research log	25%	Completion, quality, daily documentation, and responsiveness to feedback.
Reproducible project package	20%	File structure, scripts, logs, clean-session check, and saved objects.
Chemical evidence and trait design	20%	Source coverage, evidence traceability, grouping dictionary, and matrix quality.
Final report and figures	20%	Scientific logic, methods, results, figures, interpretation, and limitations.
Presentation and handoff	15%	Clear communication, final audit, continuation note, and archive readiness.

2 Instructor Review Notes and Answer Guidance

2.1 Facilitating Without Doing the Project for the Student

In dsDNA Core sessions, evidence stays at the center before answers. The strongest teaching question is often, “Which table, row, script, or citation supports that?” Students learn to diagnose problems by moving from output to source, not by guessing.

2.2 Daily Check-In Structure

This five-minute structure works at the start or end of each day:

1. What did you produce yesterday?
2. What evidence supports the main thing you think you learned?
3. What is uncertain, broken, missing, or confusing?
4. What will you produce by the end of today?
5. What decision do you need from the program lead or dsDNA Core mentor?

2.3 Expected Answers by Program Stage

Stage	Strong student answer	Needs coaching
Question design	Names the system, comparison, chemical evidence, and limitation.	Says only that they want to see what chemicals are important.
Database interpretation	Distinguishes identity, source annotation, occurrence, pathway, hazard, and assay evidence.	Treats any database hit as equivalent support.
GC-MS workflow	Explains match factor, retention time, base peak, area, and tentative identity caveats.	Says the software identified the compound without caveat.
Enrichment	Starts with validation, source coverage, and provenance.	Jumps directly to the most exciting trait or pathway.

Stage	Strong student answer	Needs coaching
Trait matrix	Explains source fields, confidence, matrix keys, and why weak traits were removed.	Uses every available column because more data seem better.
Figure interpretation	Ties each figure to a question and a source table.	Makes decorative plots without a decision or claim.
Literature integration	Uses papers to contextualize results and limitations.	Searches only for papers that support the desired conclusion.
Final report	Methods can be rerun and claims are traceable.	Reads like a narrative without reproducible evidence.

2.4 Common Program Responses

Student pattern	Program response
Overly broad project	The work is limited to one chemical list, one comparison, and one main evidence family.
Too much time cleaning messy data	The student switches to a program-reviewed case dataset for training, then returns to the real data later.
Interpreting before validation	A validation summary screenshot or table is reviewed before biological meaning is discussed.
Matrix is too wide	The matrix is narrowed to medium-confidence traits tied to the research question.
Weak writing	The student writes one claim, one evidence sentence, and one limitation sentence for each figure.
No progress after errors	The workflow moves to a small reproducible example with the exact error, input, expected output, and actual output.

2.5 Review Notes for Real-World Exercises

2.5.1 Unmatched Chemical

A good answer does not delete unmatched chemicals silently. It should preserve them in a missingness table and identify the next action: synonym search, identifier correction, manual review, or follow-up validation.

2.5.2 Hazard Code Trap

A good answer separates hazard screening from risk interpretation. Students should state that risk requires exposure, dose, route, organism, and context.

2.5.3 Pathway Does Not Mean Presence

A good answer rewrites “this pathway is active” as “this compound has KEGG pathway context consistent with...” unless independent pathway activity evidence exists.

2.5.4 Natural Product Occurrence

A good answer separates direct taxonomic evidence from broad natural product context. Students should not imply occurrence in their organism unless the source supports it.

2.5.5 Matrix Sparsity

A good answer favors a smaller matrix if it has clearer interpretation, better confidence, and sufficient coverage across compounds. Students should explain what was removed and why.

2.6 When To Slow the Program Down

The program slows down when:

- Students cannot explain their research question without reading notes.
- Source coverage is weak and students are ignoring it.
- Scripts contain absolute paths or manual steps that cannot be repeated.
- Figures are being made before variables are defined.
- Literature is being used as decoration rather than evidence.

2.7 When To Accelerate

The program accelerates when:

- The student can rerun their workflow from a clean R session.
- The grouping dictionary is reviewed and matrix outputs are stable.
- Figures have clear captions and limitations.
- The student can explain what would change their interpretation.

2.8 Final Review Questions

Before the final package is accepted, the review asks:

1. Which result is strongest, and what evidence supports it?
2. Which result is most uncertain, and why?

3. Which chemicals were excluded, unmatched, or weakly supported?
4. Which database source was most useful, and which was least useful?
5. What should the next student do first?
6. What claim should not be made from this project?

3 Bundle Maintenance and Curriculum Release

3.1 Purpose

The offline student bundle is a reproducible distribution option, not a different edition of uafR. It packages one source archive, the current student manual, training scripts, integrity metadata, installation helpers, and an acceptance test. Rebuilding the bundle for each approved package version keeps student machines aligned and makes classroom support auditable.

3.2 Release Inputs

Before Building

- Start from a reviewed repository commit with a clean working tree.
- Confirm that package tests and the release check pass without live web services.
- Build both public manuals and review their title pages, page counts, references, and institutional-mark placement.
- Run the training publication audit and offline training smoke tests.
- Confirm that no restricted data, credentials, unpublished project paths, or local usernames are present.
- Record the package version and source commit intended for the class.

3.3 Build the Public Manuals

From the repository root:

```
Rscript tools/build_training_manuals.R  
Rscript tools/check_training_publication.R
```

The build must produce:

- `training/uafR_training_manual.pdf`
- `training/uafR_instructor_guide.pdf`

The student bundle includes only the student manual. The instructor guide is a separate public companion and should not be placed in the student bundle as a hidden answer key.

3.4 Build the Student Bundle

From the repository root:

```
Rscript tools/build_student_bundle.R
```

An optional first argument chooses a different output directory:

```
Rscript tools/build_student_bundle.R /path/to/output
```

The builder regenerates the student manual, builds the package source archive, copies student-facing files, writes `MANIFEST.txt` and `CHECKSUMS.csv`, and creates a versioned zip archive.

3.5 Validate the Generated Bundle

Unzip a fresh copy rather than testing only the source folder. Then run:

```
Rscript verify_bundle_integrity.R  
Rscript preflight_check.R  
Rscript install_uafR_from_bundle.R  
Rscript run_student_acceptance_test.R
```

For a strict release check, installation should occur in a clean temporary R library on at least one supported operating system. The acceptance test remains offline by default. A live database smoke test is optional and should never determine whether a student bundle is structurally valid.

3.6 Distribution Record

Record the following before distribution:

- Package version and source commit.
- Bundle zip filename, byte size, and checksum.
- Student-manual version, revision date, page count, and checksum.
- Date and platform used for the clean installation test.
- Known warnings or temporary service limitations.
- Approved distribution location and support contact.

3.7 Updating an Existing Cohort

Issue a new versioned bundle instead of replacing files inside a distributed folder. Students should unzip the new bundle into a new directory and run `update_uafR_from_bundle.R`. Keep the old bundle until active projects have been rerun and the change is documented in each research log.

Release Boundary

Public curriculum files may describe generic workflows and simulated examples. Project-specific raw data, unpublished collaborator records, credentials, internal schedules, and restricted storage locations must remain outside the repository and outside the student bundle.

Companion Materials

The student manual, teaching scripts, worksheets, rubrics, and public package documentation are maintained in the uafR repository at <https://github.com/castrattonDSU/uafR>. Local schedules, restricted-data procedures, and project-specific access details should be supplied in a separate local supplement rather than added to the public student manual.